

## Microbiota Phylogenic Analysis Revealed Decreased Abundance of *Faecalibacterium Prausnitzii*, an Anti-Inflammatory Commensal Bacterium, in Patients with Chronic Graft-Versus-Host Disease

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### Abstract

**NF and RZG contributed equally to this work**

**Background:** Preservation of intact gut microbiota (GM) in allogeneic hematopoietic stem cell transplant (alloHCT) recipients has gained intense attention over the last decade. Loss of GM diversity and a shift in GM composition after alloHCT has been associated with adverse consequences, including increased treatment related mortality due to acute graft-versus-host disease (GVHD), infection and organ failure. Despite a growing understanding of relationships among GM, systemic inflammation and acute GVHD in early stage (up to day 100) recovery, little is known about the GM alterations in late phase (beyond 100 days) recovery of alloHCT recipients with chronic GVHD (cGVHD). We compared the GM microbiota composition (diversity, richness and taxa abundance) and prototypical cytokines (IL10, anti-inflammatory and TNF $\alpha$ , pro-inflammatory) between alloHCT recipients with and without cGVHD in late phase HCT recovery. **Method:** This cross-sectional study included two groups of adults (age > 18 years) who received T cell repleted grafts and had a fecal and blood sample collected in late phase recovery of

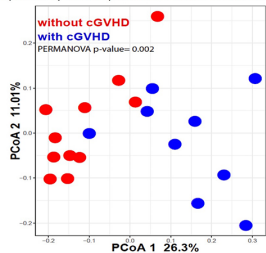
alloHCT. Group 1 (n=9) included patients with active moderate to severe cGVHD requiring systemic steroids +/- therapeutic dose Tacrolimus. Group 2 (n=11) included patients without cGVHD off Tacrolimus or on Tacrolimus taper (n=7: no cGVHD history and n=4: resolved cGVHD). Independent t-tests were used to examine clinical and transplant characteristics between groups. The V1-V3 region of the 16S rRNA gene in fecal specimens was sequenced using Illumina MiSeq. Sequenced data were processed using Quantitative Insights into Microbial Ecology (QIIME) version 1.9.1. Taxonomic assignment was done using ribosomal database project classifier with confidence cutoff set to 50%. Beta diversity analysis through Principal Coordinate Analysis (PCoA) was generated from UniFrac and Bray-Curtis distances. Difference in GM composition was tested using PERMANOVA. Alpha diversity analysis was done using Chao1 and Shannon diversity indices. Linear discriminant analysis effect size (LEfSe) was used to find markers that differentiate between groups. Cytokines were analyzed in plasma using Bioplex® multi-plex assay. Differences between groups were tested using Mann-Whitney U test. Correlations between cytokines and *F. prausnitzii* were tested using Spearman's Rho. Significance was set at  $< 0.05$  for all tests. **Results:** No significant differences noted in characteristics of the two groups except for lower performance status in cGVHD (Table 1). There was a significant difference in GM composition between groups (Figure A,  $P = 0.002$ ). Alpha diversity (richness) was significantly lower in setting of cGVHD (Figure B,  $p < 0.05$ ). Based on LEfSe analysis, 13 GM taxa were associated with Group 1 and 48 taxa were associated with Group 2. The GM community in Group 1 was enriched with Bacteroides, Lactobacillus, Clostridium and Veillonella whereas GM of Group 2 was enriched with Ruminococcus, Blautia and Faecalibacterium ( $p < 0.05$  and LDA threshold value  $> 2$ ). Interestingly, *Faecalibacterium prausnitzii* (*F. prausnitzii*) was significantly enriched in group 2 (Figure C,  $p = 0.01$ ). *F. prausnitzii*, a member of the Firmicutes phylum, is a well-known anti-inflammatory commensal bacterium. Previous studies have shown lower levels *F. prausnitzii* in patients with intestinal inflammatory diseases including active ulcerative colitis. Recently, *F. prausnitzii* has also been linked to response to anti-PD1 therapy. Differences in levels of IL10 and TNF $\alpha$  between groups were nearing significance ( $p = 0.06$  and  $p = 0.08$ , respectively). There was a significant correlation between IL10 level and *F. prausnitzii* ( $\rho = 0.55$ ,  $p = 0.01$ ) but not between TNF $\alpha$  and *F. prausnitzii* ( $\rho = -0.35$ ,  $p = 0.12$ ). **Conclusion:** These data suggest patients with cGVHD have reduced GM diversity and distinct GM composition compared to patients without cGVHD. The finding of decreased abundance of *F. prausnitzii*, in patients with cGVHD offers insights into the mechanisms of inflammation in cGVHD and suggests a role for commensal bacteria-mediated anti-inflammatory activities in this population. Future studies are needed to better understand

the immunomodulatory effects and potential therapeutic role of *F.prausnitzii* and GM composition in cGVHD to develop targeted interventions to improve outcomes of alloHCT recipients.

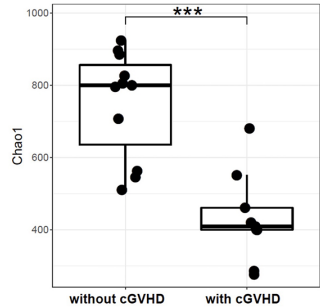
**Table 1.** Clinical and transplant characteristics of study cohorts

| Total (N = 20)                                   | With cGVHD (N= 9) | Without cGVHD (N= 11) | Difference between Groups (p value) |
|--------------------------------------------------|-------------------|-----------------------|-------------------------------------|
| Gender                                           |                   |                       |                                     |
| Male                                             | 5                 | 5                     | P=1.0                               |
| Female                                           | 4                 | 6                     |                                     |
| Median age at transplant (yrs)                   | 57 (20-69)        | 52 (23-69)            | P>0.05                              |
| Race                                             |                   |                       |                                     |
| Caucasian                                        | 8                 | 11                    | P=1.0                               |
| Other                                            | 1                 | 0                     |                                     |
| Median BMI (range)                               | 26.4 (19.9-38.8)  | 25.3 (20.3-39.2)      | P= 0.91                             |
| Median Karnofsky score (range)                   | 70.0 (60-70)      | 70.0 (60-100)         | P=0.03                              |
| Underlying disease (N)                           |                   |                       |                                     |
| AML/MDS                                          | 6                 | 6                     | P=0.66                              |
| ALL                                              | 0                 | 3                     |                                     |
| Lymphoma                                         | 2                 | 1                     |                                     |
| MPN                                              | 0                 | 1                     |                                     |
| AA                                               | 1                 | 0                     |                                     |
| Conditioning type (N)                            |                   |                       |                                     |
| RIC                                              | 5                 | 4                     | P=0.65                              |
| MA                                               | 4                 | 7                     |                                     |
| Stem cell Source                                 |                   |                       |                                     |
| PBSC                                             | 7                 | 8                     | P=1.0                               |
| BM                                               | 2                 | 3                     |                                     |
| Donor                                            |                   |                       |                                     |
| MUD                                              | 5                 | 6                     | P=1.0                               |
| MRD                                              | 3                 | 3                     |                                     |
| Haplo                                            | 1                 | 2                     |                                     |
| Median Days from transplant to enrollment (Days) | 280.0 (100-1095)  | 321 (92-1076)         | P=0.71                              |

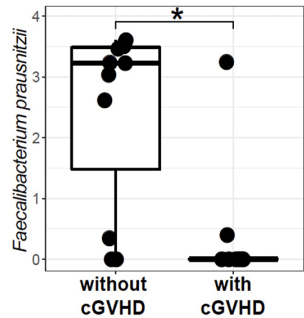
**Figure A.** Beta diversity of stool samples: OTU level Principle Coordinates Analysis (PcoA) of stool samples shows significant clustering of samples based on their group (PERMANOVA P=0.002). Percentages of variation explained by the components are shown on each axis.



**Figure B.** Alpha diversity of stool samples: There is significant reduction in richness as measured by Chao1 index (Mann–Whitney U p= 0.0002) in GM of patients with cGVHD



**Figure C.** *Faecalibacterium prausnitzii* relative abundance in stool samples: There is significant reduction in *Faecalibacterium prausnitzii* relative abundance (Mann–Whitney U p= 0.01) in cGVHD setting



Disclosures

No relevant conflicts of interest to declare.

Author notes

\* Asterisk with author names denotes non-ASH members.